A2 Solutions: Fractions, ratios, percentages and powers

← Module overview · Notes · Problems

Try every problem properly before reading its solution.

Practice

1.

(a) Common denominator 12: $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$.

(b) $\text{lcm}(6, 8) = 24$: $\frac{20}{24} - \frac{9}{24} = \frac{11}{24}$.

(c) $\frac{7}{3} - \frac{7}{4} = \frac{28}{12} - \frac{21}{12} = \frac{7}{12}$.

(d) $\frac{7}{9} \cdot \frac{15}{14} = \frac{1}{3} \cdot \frac{5}{2} = \frac{5}{6}$ (divide 7 and 14 by 7, and 15 and 9 by 3).

(e) $\frac{3}{4} \cdot \frac{5}{2} = \frac{15}{8} = 1\frac{7}{8}$.

(f) $\frac{3}{6} + \frac{2}{6} + \frac{1}{6} = \frac{6}{6} = 1$.

2.

(a) $0.375 = \frac{375}{1000} = \frac{3}{8}$ (divide by 125). $0.45 = \frac{45}{100} = \frac{9}{20}$.

(b) $\frac{5}{8} = 0.625$. $\frac{1}{7} = 0.\overline{142857} = 0.142857142857\ldots$

(c) $x = 0.\overline{27}$: $100x = 27.\overline{27}$, so $99x = 27$, $x = \frac{27}{99} = \frac{3}{11}$.

$y = 0.1\overline{6} = 0.1666\ldots$: $10y = 1.666\ldots$ and $100y = 16.666\ldots$, so $90y = 15$ and $y = \frac{15}{90} = \frac{1}{6}$.

3.

Factorise each denominator (the fractions are already in lowest terms, except $\frac{9}{75}$):

- $\frac{3}{40}$: $40 = 2^3 \cdot 5$. **Terminates** (0.075).
- $\frac{7}{12}$: $12 = 2^2 \cdot 3$, factor 3. **Recurs** ($0.58\overline{3}$).
- $\frac{9}{75} = \frac{3}{25}$: $25 = 5^2$. **Terminates** (0.12). Note: simplify first! $75 = 3 \cdot 5^2$ has a factor 3, but it cancels.
- $\frac{11}{56}$: $56 = 2^3 \cdot 7$, factor 7. **Recurs**.
- $\frac{13}{125}$: $125 = 5^3$. **Terminates** (0.104).

4.

(a) $2 + 3 + 4 = 9$ parts, each $360 \div 9 = 40$: **€80, €120, €160**.

(b)

people	4	1	7
pasta (g)	300	75	525

Divide by 4, then multiply by 7: **525 g**.

(c) $1 + 2 + 4 = 7$ parts of $35 \div 7 = 5$ kg: **5 kg cement, 10 kg sand, 20 kg gravel**.

5.

(a) $0.15 \cdot 80 = 12$.

(b) After 20% off you pay 80%: original $\cdot\, 0.8 = 64$, so the original is $64 \div 0.8 =$ **€80**.

(c) $58 \div 50 = 1.16$: an increase of **16%**.

(d) $90 \div 120 = 0.75$: a decrease of **25%**.

6.

(a) 1024 (b) -27 (c) 81 (d) -81 (e) 1 (f) $\dfrac{8}{27}$ (g) $\dfrac{1}{100} = 0.01$ (h) $\dfrac{1}{8}$ (i) 14 (j) 0.4 (k) $\dfrac{3}{5}$ (l) $3^3 = 27$

7.

(a) $72\,300\,000 = 7.23 \times 10^7$ (point moves 7 places left). $0.000\,058 = 5.8 \times 10^{-5}$ (5 places right).

(b)

(c) $2.5 \cdot 4 = 10$ and $10^6 \cdot 10^{-3} = 10^3$, so $10 \times 10^3 = 1 \times 10^4$.

(ii) $9 \div 3 = 3$ and $10^{5-(-2)} = 10^7$: 3×10^7.

(iii) $6^2 = 36$ and $(10^4)^2 = 10^8$: $36 \times 10^8 = 3.6 \times 10^9$.

8.

(a) 1 km $= 10^3$ m, so 1 km² $= 10^3 \cdot 10^3$ m² $= 10^6$ m² (a million square metres).

(b) 1 m $= 10$ dm, so 1 m³ $= 10^3$ dm³ $= 1000$ litres.

(c) 1 mm $= 10^{-3}$ m, so 1 mm² $= 10^{-6}$ m² and 5 mm² $= 5 \times 10^{-6}$ m².

School-exam level

9.

(a) $400 \cdot 1.2 = 480$, then $480 \cdot 0.8 = 384$. Final price **€384**.

(b) **No.** The overall growth factor is $1.2 \cdot 0.8 = 0.96$ in one order and $0.8 \cdot 1.2 = 0.96$ in the other. Multiplication doesn't depend on the order, so either way the price ends at 96% of €400 = €384: a 4% loss.

10.

(a) 1.03.

(b) $50\,000 \cdot 1.03^5 \approx 57\,963.7$, so about **57 964** inhabitants.

(c) We need the first n with $1.03^n \geq 2$. Trying: $1.03^{23} \approx 1.974$ and $1.03^{24} \approx 2.033$. So after **24 years**. (Population about $101\,640$.)

11.

(a) $2000 \cdot 1.03^5 \approx 2318.548\ldots$, so **€2318.55**.

(b) $5 \cdot 0.03 \cdot 2000 = 300$ of interest, so **€2300**. The €18.55 difference is "interest on interest".

12.

Price incl. BTW = price excl. BTW $\times\, 1.21$. So excl. BTW: $847 \div 1.21 =$ **€700**. BTW: $847 - 700 =$ **€147** (which is 21% of 700 ✓).

13.

(a) $6.4 \cdot 25\,000 = 160\,000$ cm $= 1600$ m $=$ **1.6 km**.

(b) Areas scale by $25\,000^2$. Easier: 1 cm on the map is 250 m $= 0.25$ km, so 1 cm² on the map is $0.25^2 = 0.0625$ km². Then 3 cm² is $3 \cdot 0.0625 =$ **0.1875 km²** (that is $187\,500$ m²).

14.

(a) $\dfrac{3.0 \times 10^8}{4.0 \times 10^7} = 0.75 \times 10^1 = 7.5$. About **7.5 times** round the Earth per second.

(b) 1 cm $= 10^{-2}$ m. $\dfrac{10^{-2}}{8 \times 10^{-5}} = \dfrac{1}{8} \times 10^3 = 125$. About **125 hairs**.

15.

Juice: $0.30 \cdot 200 + 0.10 \cdot 300 = 60 + 30 = 90$ ml, in 500 ml of mixture. $90 \div 500 = 0.18$: **18%**.

20% would be right only if equal amounts of both drinks were mixed. There is more of the 10% drink, so the mixture is closer to 10%. (The answer is a *weighted* average: $\frac{2}{5} \cdot 30\% + \frac{3}{5} \cdot 10\% = 18\%$.)

16.

(a) $\sqrt{12} = \sqrt{4 \cdot 3} = 2\sqrt{3}$ and $\sqrt{27} = \sqrt{9 \cdot 3} = 3\sqrt{3}$. Sum: $5\sqrt{3}$.

(b) $\sqrt{8} \cdot \sqrt{2} = \sqrt{16} = 4$. $\dfrac{\sqrt{72}}{\sqrt{2}} = \sqrt{36} = 6$.

(c) $8^2 = 64 < 70 < 81 = 9^2$, so $8 < \sqrt{70} < 9$. 70 is 6 above 64 and 11 below 81, so it is closer to **8**. (In fact $\sqrt{70} \approx 8.37$.)

(d) **No.** For $a = -3$: $\sqrt{(-3)^2} = \sqrt{9} = 3 \neq -3$. The square root is never negative, so $\sqrt{a^2} = a$ only when $a \geq 0$. In general $\sqrt{a^2}$ is the distance from a to 0, written $|a|$.

17.

Shop A: growth factor 0.70. Shop B: $0.80 \cdot 0.90 = 0.72$. **Shop A is cheaper**, by 2% of the original price. (Shop B's second discount is 10% of a smaller amount, so it is worth only 8% of the original.)

Challenge

18.

(a) **Lemma.** If $3 \mid a^2$ then $3 \mid a$. *Proof.* If $3 \nmid a$, then $a = 3k+1$ or $a = 3k+2$. Then $a^2 = 9k^2+6k+1 = 3(3k^2+2k)+1$ or $a^2 = 9k^2+12k+4 = 3(3k^2+4k+1)+1$. Either way a^2 leaves remainder 1, so $3 \nmid a^2$. That is the contrapositive of the lemma. □

Theorem. $\sqrt{3}$ is irrational. *Proof.* Suppose $\sqrt{3} = \dfrac{a}{b}$ in lowest terms. Then $a^2 = 3b^2$, so $3 \mid a^2$, so $3 \mid a$ by the lemma. Write $a = 3c$: then $9c^2 = 3b^2$, so $b^2 = 3c^2$, so $3 \mid b^2$ and $3 \mid b$. Then 3 divides both a and b, contradicting lowest terms. So $\sqrt{3}$ is not a fraction. □

(b) For $\sqrt{4}$ we would need: "if $4 \mid a^2$ then $4 \mid a$". That's **false**: $4 \mid 2^2$ but $4 \nmid 2$. So the step "$4 \mid a^2 \Rightarrow 4 \mid a$" fails, and the proof breaks there. Good: $\sqrt{4} = 2 = \frac{2}{1}$ is rational, so any correct proof *must* fail for it. Checking that a proof fails where the result is false is a useful habit.

19.

Argument 1 (algebra). Let $x = 0.\overline{9}$. Then $10x = 9.\overline{9}$, and subtracting, $9x = 9$, so $x = 1$.

Argument 2 (thirds). $\frac{1}{3} = 0.\overline{3}$. Multiply by 3: $1 = 0.\overline{9}$.

Argument 3 (nothing in between). If $0.\overline{9} < 1$, the number halfway between them would be bigger than $0.\overline{9}$ but less than 1. Any number less than 1 has a decimal $0.d_1d_2d_3\ldots$ with some digit less than 9 (otherwise it *is* $0.\overline{9}$). Then it is smaller than $0.\overline{9}$, which has 9 in every place. So no number lies strictly between $0.\overline{9}$ and 1, and they must be equal. (Or: the difference $1 - 0.\overline{9}$ is smaller than 0.1, than 0.01, than 0.001, …, than every positive number, so it is 0.)

20.

Let $\dfrac{a}{b}$ be in lowest terms, so $\gcd(a,b) = 1$.

If $b = 2^i5^j$: let k be the larger of i and j. Then $10^k = 2^k5^k$ is a multiple of b, say $10^k = b \cdot m$. So $\dfrac{a}{b} = \dfrac{am}{10^k}$, which is a terminating decimal (the whole number am with the point moved k places).

Only if: suppose $\dfrac{a}{b}$ terminates, so $\dfrac{a}{b} = \dfrac{N}{10^k}$ for some whole numbers N and k. Then $a \cdot 10^k = N \cdot b$, so b divides $a \cdot 10^k$. Let p be any prime factor of b. Then p divides $a \cdot 10^k = a \cdot 2^k \cdot 5^k$. By uniqueness of prime factorisation, p appears in the factorisation of a or $p = 2$ or $p = 5$. But p can't divide a, since $\gcd(a,b) = 1$. So $p = 2$ or $p = 5$. □

(Why lowest terms matters: $\frac{3}{6}$ has a 3 in the denominator but terminates, because the 3 cancels.)

21.

(a)

$$\tfrac{1}{7} = 0.\overline{142857},\ \tfrac{2}{7} = 0.\overline{285714},\ \tfrac{3}{7} = 0.\overline{428571},\ \tfrac{4}{7} = 0.\overline{571428},\ \tfrac{5}{7} = 0.\overline{714285},\ \tfrac{6}{7} = 0.\overline{857142}.$$

They all use the **same six digits in the same cyclic order**, just starting at a different place. (In the long division of $1 \div 7$, all six non-zero remainders appear; $\frac{k}{7}$ simply starts the same cycle at remainder k.)

(b) $\frac{1}{13} = 0.\overline{076923}$: period **6**. $\frac{1}{17} = 0.\overline{0588235294117647}$: period **16**.

(c) In the long division of $1 \div n$, each step has a remainder in $\{0, 1, \ldots, n-1\}$. If a remainder is 0, the decimal terminates. Otherwise every remainder is one of the $n-1$ values $1, \ldots, n-1$. Each next step depends only on the current remainder. Among the first n remainders, two must be equal (only $n-1$ possible values), and from the first repeated remainder onwards the digits repeat. So the repeating block is at most $n-1$ long.

22.

(a) $2^{300} = (2^3)^{100} = 8^{100}$ and $3^{200} = (3^2)^{100} = 9^{100}$. Since $8 < 9$, 3^{200} **is larger**.

(b) $31^{11} < 32^{11} = (2^5)^{11} = 2^{55}$, and $17^{14} > 16^{14} = (2^4)^{14} = 2^{56}$. Since $2^{55} < 2^{56}$:

$$31^{11} < 2^{55} < 2^{56} < 17^{14}.$$

So 17^{14} **is larger**.

23.

$2^{100} = (2^{10})^{10} = 1024^{10} = (1.024 \times 10^3)^{10} = 1.024^{10} \times 10^{30}$.

Since $1.024 > 1$, $1.024^{10} > 1$, so $2^{100} > 10^{30}$, which has 31 digits. If also $1.024^{10} < 10$, then $2^{100} < 10^{31}$, the smallest 32-digit number. So 2^{100} has **31 digits**.

Proof that $1.024^{10} < 1.3$ (which is far more than we need):

- $1.024^2 = 1.048576 < 1.05$;
- so $1.024^4 < 1.05^2 = 1.1025$;
- so $1.024^8 < 1.1025^2 = 1.21550625 < 1.22$;
- so $1.024^{10} = 1.024^8 \cdot 1.024^2 < 1.22 \cdot 1.05 = 1.281 < 1.3$.

(Calculator: $1.024^{10} \approx 1.268$, so $2^{100} \approx 1.268 \times 10^{30}$.)

24.

Growth factors: $\left(1 + \frac{p}{100}\right)\left(1 - \frac{p}{100}\right) = 0.96$. Write $q = \frac{p}{100}$. Then $(1+q)(1-q) = 1-q^2$ (expand the brackets: $1 - q + q - q^2$). So $1 - q^2 = 0.96$, $q^2 = 0.04$, $q = 0.2$ (as $q > 0$). So $p = \mathbf{20}$.

Check: $1.2 \cdot 0.8 = 0.96$. ✓ (This is the same as problem 9.)

25.

(a) Common denominator $n(n+1)$:

$$\frac{1}{n+1}+\frac{1}{n(n+1)}=\frac{n}{n(n+1)}+\frac{1}{n(n+1)}=\frac{n+1}{n(n+1)}=\frac{1}{n}. \quad \square$$

(b) $\frac{2}{7}=\frac{1}{7}+\frac{1}{7}$, but the fractions must be different. Use (a) on one of them: $\frac{1}{7}=\frac{1}{8}+\frac{1}{56}$, giving $\frac{1}{7}+\frac{1}{8}+\frac{1}{56}$, which has three terms. For two terms, try $\frac{2}{7}-\frac{1}{4}=\frac{8-7}{28}=\frac{1}{28}$. So $\dfrac{2}{7}=\dfrac{1}{4}+\dfrac{1}{28}$.

(How to find it: the largest unit fraction below $\frac{2}{7}\approx 0.286$ is $\frac{1}{4}$; see what is left. This is the *greedy method*.)

(c) Start from $1=\frac{1}{2}+\frac{1}{3}+\frac{1}{6}$ (problem 1f), then split $\frac{1}{6}=\frac{1}{7}+\frac{1}{42}$ using (a):

$$1=\frac{1}{2}+\frac{1}{3}+\frac{1}{7}+\frac{1}{42}.$$

Another answer: $\frac{1}{2}+\frac{1}{4}+\frac{1}{6}+\frac{1}{12}=\frac{6+3+2+1}{12}=1$.

26.

(a) Mediant: $\dfrac{1+2}{2+3}=\dfrac{3}{5}=0.6$, and $\frac{1}{2}=0.5<0.6<0.666\ldots=\frac{2}{3}$. ✓

(b) All of a, b, c, d are positive, so we can multiply inequalities by denominators without changing direction. $\dfrac{a}{b}<\dfrac{c}{d}$ means $ad<bc$.

Left inequality: $\dfrac{a}{b}<\dfrac{a+c}{b+d}$ means $a(b+d)<b(a+c)$, that is $ab+ad<ab+bc$, that is $ad<bc$. True.

Right inequality: $\dfrac{a+c}{b+d}<\dfrac{c}{d}$ means $d(a+c)<c(b+d)$, that is $ad+cd<bc+cd$, that is $ad<bc$. True. $\square$

(Picture: think of $\frac{a}{b}$ as a slope. Adding the vectors (b,a) and (d,c) gives a slope between the two.)

(c) Given two fractions $x<y$, the mediant m_1 is a fraction strictly between them. Then the mediant of x and m_1 is a fraction strictly between x and m_1, and so on. This gives an endless list $x<\cdots<m_3<m_2<m_1<y$ of different fractions between x and y. So there are infinitely many. (Simpler but less elegant: $\frac{x+y}{2}$ also works.)